

A non-operator movement approach to Geis-ambiguities*

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Abstract: The primary goal of this paper is to argue against the conventional operator movement approach to Geis-ambiguities. The motivation for favoring a non-operator movement account consists of two empirical findings. On the one hand, the *wh*-element *when* that introduces temporal adverbial clauses is better analyzed as a base-generated *wh*-complementizer rather than a mobile *wh*-operator. On the other hand, the ambiguous interpretations of adverbial *when* have some close correlation with different uses of specific bridge verbs (i.e., *SAY* verbs). Based on these observations, this paper proposes an alternative approach that appeals to feature valuation and Agree instead of any kind of movement.

Keywords: Geis-ambiguities, temporal adverbial clauses, operator movement, *wh*-complementizer, *SAY* verbs, event argument

1 Introduction

In this short paper, I present a number of novel findings concerning *Geis-ambiguities* (Geis 1970), which lead to an account that is different from the conventional, operator movement-based approach (Larson 1987, 1990, Haegeman 2007, 2009, 2010, 2012, a.o.). The phenomenon is also known as *ambiguity in high/low construal* in the relevant literature: as exemplified in (1), the *wh*-word *when* in a temporal adverbial clause can relate to either the local verb *say* or a more deeply-embedded one *leave*, and give rise to two different interpretations. The high construal of *when* refers to the time that Puffy made a certain statement, viz., that she would leave (sometime) (1a), whereas its low construal refers to some time that Puffy said would be the time of her presumed departure (1b).

- (1) I saw Puffy in Canary Wharf *when* [she said [that she would leave]].
- a. High construal: at the time that Puffy made the statement of departure
 - b. Low construal: at the time of Puffy's presumed departure

Geis's original analysis treats temporal adverbial clauses like (1) as restrictive relative clauses, the antecedents of which are deleted, thereby assuming that "*when* is derived from some time adverbial that moves to clause-initial position" (Geis 1970:77). Based on Geis's idea, Larson (1987, 1990) proposes that *when* introduces a free relative clause, and that the ambiguous readings correspond to structures in which the *wh*-word originates in the upper or the lower adverbial clause. As schematized in (2a), *when* is postulated to be first base-generated within the higher TP-domain of the temporal adverbial clause and then

*Acknowledgement to be inserted

move to its left periphery, leading to the high construal of the *wh*-word. By contrast, the low construal of *when* derives from a different configuration where it starts long-distance movement from the lower TP-domain of the adverbial clause, as illustrated in (2b).

- (2) I saw Puffy in Canary Wharf [_{CP} *when* [_{TP} she said [_{CP} that [_{TP} she would leave]]]].
- a. High construal: [_{CP} *when*_i [_{TP} she said [_{CP} that [_{TP} she would leave]] *t*_i]]
 - b. Low construal: [_{CP} *when*_i [_{TP} she said [_{CP} *t*_i that [_{TP} she would leave *t*_i]]]]

Following Larson (1987, 1990) and many others, Haegeman (2007, 2009, 2010, 2012) adopts the analysis that temporal adverbial clauses are derived by *wh*-movement of an operator (e.g., *when*) to the left periphery. She argues that the very fact that in English, *when*, as the ‘conjunction’ introducing temporal adverbials (3a), is isomorphic to an interrogative operator (3b) and to a relative operator (3c) supports a uniform movement analysis of *when*. She also points out that, in many other languages, the adverbial ‘conjunction’ has the same multiple behavior as English *when*, formally corresponding to either an interrogative element (e.g., French *quand*, Italian *quando*, Spanish *cuando*) or a relative element (e.g., Norwegian *når*).

- (3)
- a. I was still sleeping *when* Puffy left. (adverbial *when*)
 - b. *When* did Puffy leave? (interrogative *when*)
 - c. I remember (the day) *when* Puffy left. (relative *when*)

The operator movement approach sketched out above captures the ambiguous interpretations of adverbial *when* in a pretty straightforward way. However, this paper offers two empirical arguments that pose challenges to this analysis. First, since adverbial *when* is argued to be isomorphic to its interrogative counterpart at least, which is standardly regarded as an operator involved in *wh*-movement,¹ a reasonable prediction is that adverbial *when* should behave identically to interrogative *when* in all respects. This is not borne out, as will be seen in Sect. 2.

The second challenge comes from recent work by Yan (2022), where Geis-ambiguities are reported missing in some specific configuration. Considering (1) again, *say* in the upper adverbial clause is classified into *bridge verbs* according to Erteschik-Shir (1973), since it allows long-distance extraction out of the embedded clause. Nonetheless, if the same temporal adverbial clause involves a non-bridge verb, such as *exclaim* (4), only the high construal is available: *when* can only refer to the time that Puffy made the exclamation (4a); the analogous low construal to (4b) is unavailable.

- (4) I saw Puffy in Canary Wharf *when* [she exclaimed [that she would leave]].
- a. High construal: at the time that Puffy made the exclamation
 - b. *Low construal: at the time of Puffy’s presumed departure

¹ By contrast, the analysis of the relative *wh*-operator varies and requires precise formal implementation, especially in the case of free relatives, where the *wh*-element is argued to be either in the C head or in the Spec,CP. See, e.g., Citko (2002) for relevant discussion and references therein. I will not touch upon the relative *wh*-operator until Sect. 2.5, where I briefly enumerate several diagnostics in favor of the head account for it.

Though it seems attractive to account for the disappearing low construal of *when* via the idea that movement of *when* from the lower adverbial clause is blocked by *exclaim*, it is unclear what in effect prevents this long-distance extraction. The *prima facie* blocking effect is reminiscent of island effects, but it still remains mysterious what kind of islands *exclaim* constitutes.

Taking these observations into account, I argue against the conventional movement approach to Geis-ambiguities in this paper. Instead, I propose a different approach that involves no operator movement at all. Based on the same empirical findings mentioned above, I argue that (i) adverbial *when* is a base-generated *wh*-complementizer rather than a *wh*-operator that moves to the left periphery, and that (ii) ambiguous high/low readings are reflections of the distinct eventive/stative uses of SAY verbs (as proposed in Grimshaw 2015 and Major 2021). To link these two pieces together, I adapt the feature valuation system from Adger and Ramchand (2005), by which I show how the ambiguous interpretations of adverbial *when* are accounted for without appealing to movement.

The paper is structured as follows. Section 2 present a number of empirical findings that argue against the operator status of adverbial *when*. Section 3 first shows the correlation of Geis-ambiguities with the two uses of SAY verbs, based on which my alternative approach to the same ambiguous effects is laid out then. Section 4 concludes the paper.

2 Adverbial *when* as *wh*-complementizer

In this section, I argue that adverbial *when* is better understood as a *wh*-complementizer, which is essentially distinct from the archetypal *wh*-operator, i.e., interrogative *when*. Compared to a *wh*-operator (which canonically occupies the Spec,CP), a *wh*-complementizer is base-generated in the C head instead of being moved from some other position (cf. Bresnan and Grimshaw 1978). All the differences between adverbial and interrogative *when* can be attributed to the characteristics of the former being as a *wh*-complementizer. The result of comparison between the two kinds of *wh*-elements also suggests that relative *when* be analyzed as a base-generated *wh*-complementizer.

2.1 *Wh*-adverb float

The first distinction between adverbial and interrogative *when* comes from their co-occurrence with floating *wh*-adverbs. As pointed out by McCloskey (2000:63, fn. 8), in Standard English, adverbs like *exactly* and *precisely* can not only form a constituent with *wh*-phrases, but also be separated from *wh*-phrases with which it is construed. For instance, the temporal expression *at 9 p.m.* is modified by the adverb *exactly* in (5), implying that these two elements form a single constituent together. Turning to questions, we find that interrogative *when* and *exactly* can show up together in the topmost Spec,CP (5a). This follows naturally the idea that the *wh*-operator and the *wh*-adverb are actually one constituent in the base position, and the former pied-pipes the latter to its final landing site in the matrix clause.

- (5) Piet said (that) he was drawing the picture *at 9 p.m. exactly*.
 a. *When_i exactly* did Piet say (that) he was drawing the picture *t_i*?

- b. *When_i* did Piet say *exactly* (that) he was drawing the picture *t_i*?
- c. *When_i* did Piet say (that) he was drawing the picture *exactly*?

Interrogative *when* is also able to occur alone in the left periphery with *exactly* being stranded either in the intermediate Spec,CP (5b) or in the base position (5c). We do not need to concern ourselves with the syntactic mechanism of floating *wh*-adverbs here (see McCloskey 2000 for his original solution). What matters to our discussion below is that interrogative *when*, as a *wh*-operator, is capable of pied-piping and stranding *wh*-adverbs via successive-cyclic movement in the derivation of temporal interrogative clauses.

By contrast, adverbial *when* differs from its interrogative counterpart in being incompatible with *exactly* under any circumstances, as seen in (6): adverbial *when* can neither pied-pipe the *wh*-adverb to the left periphery (6a), nor strand it in some lower positions in the midst of derivation (6b)/(6c). The contrast between (5) and (6) is unexpected under the view that both temporal interrogative and adverbial clauses are derived by the same mechanism, namely, *wh*-operator movement. In other words, what the contrast suggests is that adverbial clauses involve a kind of derivation that is different from interrogative ones.

- (6) *When* Piet said (that) he was drawing the picture, he recalled his best friend.
 - a. **When exactly* Piet said (that) he was drawing the picture ...
 - b. **When* Piet said *exactly* (that) he was drawing the picture ...
 - c. **When* Piet said (that) he was drawing the picture *exactly* ...

One way of capturing this difference directly is to argue that adverbial *when* is not the same *wh*-operator as interrogative *when* (cf. Bresnan and Grimshaw 1978). Therefore, I presume that adverbial *when* is in effect a *wh*-complementizer that is base-generated in the C head position. This idea kills two birds with one stone. First of all, it captures the distribution of adverbial *when* directly, that is, it occupies the highest projection of the clausal architecture and heads a type of subordinate clauses. Secondly, all the ungrammatical examples in (6) get explained right away: since adverbial *when* is directly base-generated in the C head, it never participates into a movement chain that starts from the embedded clause and moves through its edge; thus, it is impossible for adverbial *when* to pied-pipe and strand the *wh*-adverbs in the first place, as shown in (6b) and (6c). With respect to (6a), I suppose that it is due to the reason that adverbial *when*, as a C head, cannot be modified by the *wh*-adverbs, unlike interrogative *when* which has a phrasal-status and can form a constituent with the same adverbs. The distinct positions taken by interrogative and adverbial *when* are schematized respectively in (7) and (8).²

- (7) Interrogative *when* in the Spec,CP
 [CP *when*_{interrogative} [C' C ...]]
- (8) Adverbial *when* in the C head
 [CP [C' *when*_{adverbial} ...]]

2.2 *Wh-the-hell* phrases

The second argument favoring adverbial *when* as a *wh*-complementizer arises from *wh*-the-hell phrases. Pesetsky (1987) discusses various situations with respect to ‘aggressively non-D-linked *wh*-phrases’, e.g., *who the hell*. At least in English, to form a *wh*-the-hell question, the *wh*-element must (i) undergo movement in syntax, and (ii) not be linked to the discourse (i.e., it is non-D-linked).

The first requirement is exemplified by the difference between root questions in (9) and echo questions in (10). In all these examples a non-D-linked interrogative *when* is involved. When this *wh*-element moves to the left periphery, it forms a root question about the temporal modification (9a). The ex-situ *when* can be further modified by *the hell* to “express surprise in the answer” (Pesetsky 1987:111), as given in (9b). By comparison, the same *wh*-element is also able to stay in-situ and forms an echo question (10a). Nonetheless, the in-situ *when* is incompatible with the modification of *the hell* (10b), unlike its ex-situ counterpart (9b). Without any other assumptions, the only difference between these two configurations is that interrogative *when* moves overtly in syntax in the derivation of root questions but not in that of echo questions. Syntactic movement, thus, offers a direct account for the distinct results seen above: *wh*-ex-situ is compatible with the modifying *the hell*, whereas *wh*-in-situ is not.

(9) Root question

- a. *When_i* did Piet draw the picture *t_i*?
- b. *When_i the hell* did Piet draw the picture *t_i*?

(10) Echo question

- a. Piet drew the picture *when*?
- b. *Piet drew the picture *when the hell*?

The second requirement is demonstrated by examples in (11), where the D-linked *which*-phrases are

² Note that in Haegeman (2009:386-387, fn. 2), she also brings up the possibility that adverbial *when* is a genuine complementizer and spells-out a C head. But Haegeman still assumes that there is a null operator that moves from the TP-domain into the Spec,CP, as illustrated in (ii).

- (i) Adverbial *when* in the C head with a null operator in the Spec,CP
 [CP OP_i [_{C'} *when*_{adverbial} [TP ...*t_i* ...]]]

It is worth discussing whether the existence of the null operator is required. If a null operator was involved in the derivation of temporal adverbial clauses like (6) and could be modified by *wh*-adverbs, one would need to assume that, unlike a *wh*-operator, a null operator is unable to pied-pipe the *wh*-adverbs due to some independent reason (cf. Wu 2022), thereby accounting for the ungrammatical configurations (6a) and (6b). Then one would leave (6c) unexplained, because in this configuration, the null operator does not pied-pipe the adverb at all: if only the null operator itself moved to the left periphery, the derivation was predicted acceptable in theory, contrary to the fact. One complementary assumption would be that the null operator is also unable to strand the *wh*-adverbs in the base position. If so, it seems redundant to assume the existence of the null operator in the first place, since the same results can be captured in a more economical way, as outlined in Sect. 2.1. Alternatively, one could assume at the outset that a null operator could not be modified by *wh*-adverbs at all. Then that all the examples in (6) are ungrammatical would be predicted, since *wh*-adverbs would not be able to modify any constituents in this configuration. However, what is the motivation for assuming such an unmodifiable null operator is still unclear to me. Therefore, I will no longer entertain such an analysis in the rest of the paper.

involved. Imagine a context where both participants in a conversation know that Piet has several habitual working time slots every day (for instance, 12 p.m., 3 p.m., 6 p.m., etc.). Within such a context, (11a) is a natural question to raise as *which time* entails some linking to the discourse. By contrast, (11b) is ungrammatical since there exists some pragmatic conflict between the aggressively D-linked phrase *which time* and the aggressively non-D-linked modifier *the hell* (Pesetsky 1987). Note that the ungrammaticality of (11b) cannot be attributed to the first requirement, because *which time* shows up in the left periphery, which I take as evidence that it already undergoes movement in syntax.

- (11) a. *Which time_i* did Piet draw the picture *t_i*?
 b. **Which the hell time* did Piet draw the picture *t_i*?

With both requirements in mind, we can now use them as diagnostics to test the characteristics of adverbial *when* such as (12a). As shown in (12b), adverbial *when* resists the modification of the non-D-linked *the hell*, in contrast to its interrogative counterpart in the root question (9b). Along the uniform operator movement analysis, we again encounter an unexpected result: if interrogative and adverbial *when* were the same kind of *wh*-operator, we would predict that they behave identically with respect to the modification of *the hell*, unlike the contrast we have seen between (9b) and (12b).

- (12) a. *When* Piet drew the picture, he recalled his best friend.
 b. **When the hell* Piet drew the picture, he recalled his best friend.

Since there is no motivation to suppose that adverbial *when* in (12) is D-linked, the remaining method of explaining the ungrammatical (12b) is to presume that adverbial *when* does not undergo movement in syntax, which fits in with my viewpoint that this *wh*-element is base-generated in the C head position as a *wh*-complementizer, rather than a *wh*-operator that moves to the Spec,CP.

2.3 Pair-list reading

The third difference between the two kinds of *wh*-elements is concerned with their semantic interpretations. Consider (13) first where interrogative *when* forms a root question. The interpretation of interrogative *when* is ambiguous: it has either an individual reading (13a), calling for answers that refer to a single time that every boy woke up simultaneously, or a pair-list reading (13b), namely, for every boy, they woke up at their own time. As argued by Chierchia (1993) and Xiang (2023), since this kind of pair-list reading of interrogatives only arises in questions with a quantificational subject, which involves ‘quantification into questions’ (unlike pair-list multiple *wh*-questions), (13b) can also be termed a QIQ reading.

- (13) *When* did every boy wake up?
 a. ‘When is the time *t* such that every boy woke up at *t*?’ (individual reading)
 ‘Every boy woke up at 6 a.m.’
 b. ‘For every boy *x*, when is the time *t* such that *x* woke up at *t*?’ (pair-list reading)

‘Piet woke up at 6 a.m., Wassily woke up at 7 a.m., Pablo woke up at 8 a.m.’

One prominent approach to explaining such ambiguities is due to May (1985, 1988), who proposes that an ambiguous sentence like (13) has the LF representation in (14). The *wh*-operator, which moves in syntax, and the quantificational subject, which undergoes QR at LF, can be interpreted freely regarding scope relative to each other, thus leading to the scope ambiguities as well as the corresponding interpretations of interrogative *when*. As pointed out by Williams 1988 and many others, May’s account is certainly not unproblematic, but his proposed LF representation (14) still lays the foundations for a lot of other alternative accounts, such as Chierchia (1993) and Xiang (2023). What is common to both Chierchia’s analysis and Xiang’s is that, for the pair-list reading to be possible, the trace of the quantificational subject has to c-command the trace of the *wh*-operator. Let us tentatively accept that this structural relation at LF is on the right track and is indeed the key to the pair-list/QIQ reading of interrogative *when*.

- (14) The LF representation of (13)
- [*when*_{*i*} [every boy_{*j*} [*t*_{*j*} woke up *t*_{*i*}]]]
-

Now we can look at (15), a different temporal clause introduced by adverbial *when*. Unlike its ambiguous interrogative counterpart in (13), the individual reading of adverbial *when* (15a) is clearly more accessible than its pair-list (or QIQ) reading (15b).³ Why is this pair-list reading relatively degraded? Once again, we come across a situation where the result of comparison (i.e., (13) vs. (15)) is not expected under the *wh*-operator analysis of adverbial *when*, but it receives a straightforward explanation under the *wh*-complementizer analysis.

- (15) *When* every boy woke up, the sun was shining.
- a. There is a time *t* such that every boy woke up at *t*. (individual reading)
- b. ?For every boy *x*, there is a time *t* such that *x* woke up at *t*. (?pair-list reading)

Treating adverbial *when* as a base-generated *wh*-complementizer gives us the LF representation of (15) in (16). We still assume that the quantificational subject undergoes QR at LF, leaving a co-indexed trace behind. However, at this LF there is no other trace c-commanded by that of the quantificational subject, since adverbial *when* is base-generated in its C head position directly without undergoing any movement in syntax. The pair-list reading of adverbial *when* (15b), thus, cannot be legitimately represented by this LF. But the surface scope of this *wh*-element over the quantificational subject still holds, which renders the individual reading (15a) the only possible one.

- (16) The LF representation of (15)

³ The judgment in (15) is shared by my four consultants, all of whom are native speakers of British English.

[when [every boy_j [t_j woke up ...]]]

2.4 Cross-linguistic evidence

The previous three arguments have shown that adverbial *when* and interrogative *when* behave differently in syntax, though they look alike regarding their surface forms. As mentioned in Sect. 1, the identical forms of these two *wh*-elements across several languages are a crucial argument for Haegeman (2007, 2009, 2010, 2012) to analyze adverbial *when* as a mobile *wh*-operator. If my proposed analysis was on the right track, we would expect to see cross-linguistic evidence that clearly displays a lexical distinction between different *wh*-elements. Fortunately, we do find some of them in languages that are closely-related to English.⁴

One example is Scottish Gaelic, of which *when* is realized by two different lexical items.⁵ As shown in (17a), *nuair* is the word that functions as adverbial *when* in Scottish Gaelic. It is also used to introduce temporal relative clauses, as given in (17b). By comparison, interrogative *when* in Gaelic is spelled out by a different lexical item *cuine*, no matter in a root question (17c) or an embedded one (17d). The same results are also found in other two Celtic languages, namely, Irish and Welsh. In Irish, adverbial and relative *when* use the same word *nuair* as with Scottish Gaelic, whereas interrogative *when* changes to a different one *cathain*. In Welsh, the distinction between adverbial/relative *when* and interrogative *when* still holds: the former is realized by *pan*, while the latter is by *pryd*.

(17) *When* in Scottish Gaelic

- a. Bidh Sileas ann, *nuair* a bhios mise ann.
be.FUT Julia there when that be.FUT.REL I.EMPH there
'Julia will be there, when I will be there.' (adverbial *when*)
- b. Tha cuimhne agam air an latha *nuair* a bha sinn ann.
be.PRS memory at.me on the day when that be.PST we there
'I remember the day when we were there.' (relative *when*)
- c. *Cuine* a bhios Sileas ann?
when that be.FUT.REL Julia there
'When will Julia be there?' (non-embedded interrogative *when*)
- d. Dh'fhaighnich mi *cuine* a bhios Sileas ann.
ask.PST I when that be.FUT.REL Julia there
'I asked when Julia will be there.' (embedded interrogative *when*)

Another example is Icelandic, whose morphosyntactic behavior regarding different kinds of *when* represents a few Germanic languages. As seen in (18a) and (18b), temporal adverbial and relative clauses are introduced by the same word *þegar*, which also means 'then' in other cases. In contrast, temporal interrogative

⁴ I thank two of my consultants for the data presented here: Consultant 1 for Scottish Gaelic and Consultant 2 for Icelandic. I also thank my other consultants for offering me the data that are only mentioned here: Consultant 3 for Irish, Consultant 4 for Welsh, Consultant 5 for Swiss German and German, and Consultant 6 for Dutch.

⁵ Abbreviations: EMPH = emphatic, FUT = future, PL = plural, PRS = present, PST = past, REL = relative, SBJV = subjunctive, SG = singular

ative clauses are introduced by a different item *hvenær*, as exemplified in (18c) and (18d). The dichotomy between adverbial/relative *when* and interrogative *when* exists in German, Swiss German, and Dutch as well. But the lexical selection of adverbial *when* in these three languages is a bit trickier than that in Icelandic: all of them have two lexical items that can act as adverbial/relative *when*, the choice of which seems to be subject to some other properties of clauses, e.g., tense. I will leave detailed discussion of these three languages for another occasion.

(18) *When* in Icelandic

- a. Júlía verður þarna þegar ég verð þarna.
 Julia will.be.3SG.PRS there when I will.be.1SG.PRS there
 ‘Julia will be there, when I will be there.’ (adverbial *when*)
- b. Ég man eftir deginum þegar við vorum þarna.
 I remember.1SG.PRS after day.the when we be.1PL.PST there
 ‘I remember the day when we were there.’ (relative *when*)
- c. *Hvenær* verður Júlía þarna?
 when will.be.3SG.PRS Julia there
 ‘When will Julia be there?’ (non-embedded interrogative *when*)
- d. Ég spurði *hvenær* Júlía yrði þarna.
 I ask.1SG.PST when Julia will.be.3SG.PST.SBJV there
 ‘I asked when Julia would be there.’ (embedded interrogative *when*)

To sum up, even though cross-linguistic evidence presented above does not suffice to show directly that adverbial *when* in English is a base-generated complementizer, it still supports my idea that the syntax of adverbial *when* should be considered distinct from that of interrogative *when*.

2.5 Note on relative *when*

Before wrapping up this whole section, I would like to make a quick note on relative *when*. As already mentioned in footnote 1, whether the relative *wh*-operator occupies the Spec,CP or the C head remains controversial in the literature (Citko 2002), which is why I put relative *when* aside in the previous discussion. Cross-linguistic evidence in Sect. 2.4 shows that, at least in Scottish Gaelic and Icelandic, relative and adverbial *when* are realized by the same lexical item, which implies that relative *when* might behave identically to its adverbial counterpart, namely, a *wh*-complementizer.

Turning the three phenomena discussed above into diagnostics and applying them to relative *when*, we find that relative *when* indeed behaves like a *wh*-complementizer. That is, neither is relative *when* able to pied-pipe and strand the *wh*-adverb *exactly* (19), nor able to co-occur with the aggressively non-D-linked modifier *the hell* (20). In addition, when relative *when* interacts with a quantificational subject (21), its individual reading is also more accessible than its pair-list one. These findings can provide further support for the head account of relative *when*, a tangential topic that I will not continue in the rest of the paper.

- (19) I remember (the time) *when* Piet said (that) he was drawing the picture.

- a. *I remember (the time) *when exactly* Piet said (that) he was drawing the picture.
 - b. *I remember (the time) *when* Piet said *exactly* (that) he was drawing the picture.
 - c. *I remember (the time) *when* Piet said (that) he was drawing the picture *exactly*.
- (20) *I remember (the time) *when the hell* Piet drew the picture.
- (21) I remember (the time) *when* every boy woke up.
- a. I remember that there is a time t s.t. every boy woke up at t . (individual reading)
 - b. ?I remember that for every boy x , there is a time t s.t. x woke up at t . (?pair-list reading)

Coming back to adverbial *when*, I have shown that it differs from interrogative *when* in a number of respects, and that their distinctions can be accounted for straightforwardly once we adopt the idea that adverbial *when* is a base-generated *wh*-complementizer, whereas interrogative *when* is a mobile *wh*-operator. One significant consequence of this line of thinking is that, the conventional, operator movement-based approach to Geis-ambiguities can no longer hold. A different approach that does not appeal to movement is called for, which will be laid out in the following section.

3 An alternative account

In this section I propose a non-operator movement approach to Geis-ambiguities. The approach is motivated by the observation documented in Yan (2022) that ambiguous interpretations of adverbial *when* correlate with syntactic and semantic differences of *SAY* verbs (in the sense of Grimshaw 2015). Following the proposal made by Major (2021), I attribute these differences to the outmost VP-shell of the entire structure of *SAY* verbs. For those that are eventive in nature, I postulate that a Davidsonian event argument (see Davidson 1967) is located in the specifier of their outmost VP-shell (i.e., Spec,VoiceP) (cf. Kratzer 1995, 1996, Stowell 1996, 2007). I propose that the interpretation of adverbial *when* is determined by undergoing Agree with this event argument, and I adapt the feature valuation system from Adger and Ramchand (2005) to implement such a dependency.

Before delving into any detail, it is necessary to delimit in advance the range of empirical phenomena which my proposed analysis is built upon. Our introductory example of Geis-ambiguities (1), repeated in (22) below, utilizes *say* as the matrix verb of the temporal adverbial clause. In addition to *say*, other verbs that are frequently used to demonstrate the same ambiguous effects in the literature include *tell* (e.g., Geis 1970:77, (36)) and *claim* (e.g., Larson 1990:170, (3)).

- (22) I saw Puffy in Canary Wharf when [she *said* [that she would leave]]. (= (1))
- a. High construal: at the time that Puffy made the statement of departure
 - b. Low construal: at the time of Puffy's presumed departure

Though the list is definitely not exhaustive, the three bridge verbs mentioned above, plus the non-bridge

one *exclaim* (4) (repeated in (23)), which shows the lack of low construal of adverbial *when*, all pertain to the manner of *saying*, and thus can be grouped together as SAY verbs, as proposed in Grimshaw (2015). These verbs constitute the foundation of our further discussion of Geis-ambiguities in the rest of the paper. For other verbs that might show similar ambiguities but cannot be easily grouped into the SAY schema, I will leave them for future research.

(23) I saw Puffy in Canary Wharf when [she *exclaimed* [that she would leave]]. (= (4))

- a. High construal: at the time that Puffy made the exclamation
- b. *Low construal: at the time of Puffy's presumed departure

3.1 Two uses of *say* in Major (2021)

I begin with Major (2021)'s idea that there are two uses of *say* in English and his syntactic analysis of their differences. Building on the SAY schema proposed by Grimshaw (2015), Major argues that *say* alternates between being an eventive verb and a stative verb, as shown in (24).

(24) I visited the suspects in the detention center yesterday.

- a. Suspect #2 *said* that he is guilty. (Agent, eventive *say*)
- b. Suspect #2's sweating *says* that he is guilty. (Source, stative *say*)

(Adapted from Major 2021:39, (60))

(24a) is an instance of eventive *say*, where the most natural reading is that a communicative or speech event happened at some point in the past. The subject *Suspect #2* is interpreted as the Agent of the event, who engages in the physical production of speech. Stative *say*, as in (24b), does not refer to an actual communicative act; instead, its most natural reading is that some speech content (i.e., 'he is guilty') is stated (or conveyed) by the subject *Suspect #2's sweating*, which is assumed to hold at the present time.⁶ The former is defined as a 'Linguistic Material' (LM) argument, and the latter is understood as the Source of this LM (see Grimshaw 2015 and Major 2021 for more fine-grained discussion).

In addition to the semantic distinctions seen above, Major shows that the syntax of the two uses of *say* is quite different as well. For instance, eventive *say* is compatible with subject-oriented adverbs (e.g., *enthusiastically*) and manner adverbs (e.g., *loudly*) (25a). Besides, the eventive use of *say* is also compatible with the progressive aspect (25b). In contrast, stative *say* shows the opposite pattern: it is incompatible with either subject-oriented/manner modification (26a) or the progressive aspect (26b).

(25) I visited the suspects in the detention center yesterday. (eventive *say*)

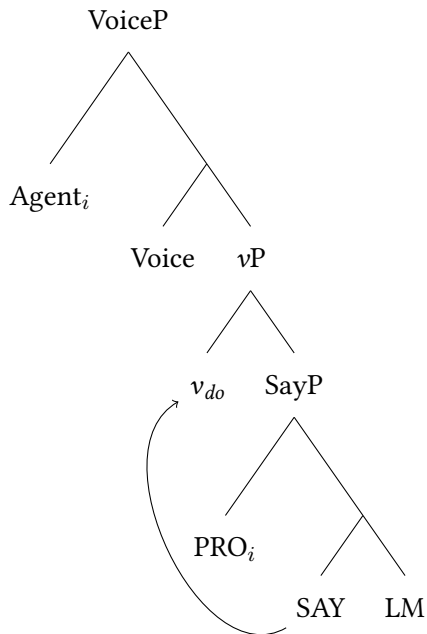
- a. Suspect #2 *enthusiastically/loudly said* that he is guilty.

⁶ Hazel Pearson (pers.comm.) points out to me that the term 'stative' is not very clear, since this concept can be decomposed into 'habitual' or 'generic'. Given the main purpose of this work is to understand the mechanism behind the ambiguous interpretations of adverbial *when*, I leave aside the issue of defining 'stative *say*' precisely, and simply use it as a cover term for *say* or (SAY verbs) that are non-eventive.

- b. Suspect #2 *was saying* that he is guilty.
- (26) I visited the suspects in the detention center yesterday. (stative *say*)
- a. *Suspect #2's sweating *enthusiastically/loudly says* that he is guilty.
- b. *Suspect #2's sweating *is saying* that he is guilty.

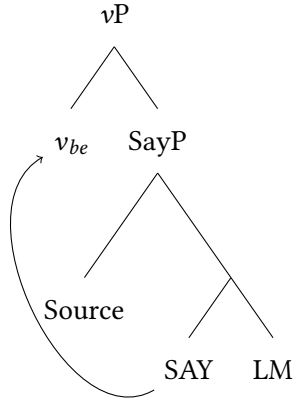
Major ascribes these differences to the underlying structures of eventive and stative *say*. Following Grimshaw (2015), he posits that both structures contain a projection headed by the light verb SAY as the lowest VP-shell, in which the LM argument is taken as complement to SAY, and the subject is base-generated in the Spec,SayP. On the basis of the “flavours of little *v*” idea (Folli and Harley 2005), he proposes that the fundamental difference between the two uses of *say* lies in their *v* head that merges with SayP. For eventive *say* (27), SayP is selected by v_{do} , which is further selected by Voice (a VoiceP/*v*P-splitting idea as proposed in Pylkkänen 2008). The subject of SayP in this configuration is assumed to be PRO, which is obligatorily controlled by the external argument (i.e., the Agent) introduced in the Spec,VoiceP.

(27) Structure of eventive *say*



As to stative *say* (28), SayP is selected by v_{be} instead of v_{do} . Here *v*P is assumed to be the highest VP-shell of the entire structure of stative *say*, that is, it is not further selected by the Voice head. Due to the lack of VoiceP, whose specifier position is canonically where the external argument is base-generated, the Source argument is directly introduced in the Spec,VoiceP. In both structures (27) and (28), *say* is spelled out via the movement of SAY to *v*.

(28) Structure of stative *say*



Major (2021:54) claims that it is the presence of VoiceP and v_{do} that permits the cluster of properties that eventive *say* exhibits in (25), compared to how stative *say* behaves in (26). I will return to these two structures of *say* when I lay out my alternative approach to Geis-ambiguities and its implementation in Sect. 3.3 and 3.4.

3.2 Correlation between SAY verbs and ambiguities of *when*

In this subsection I show that the contrast between the two uses of *say* seemingly correlates with ambiguity in high/low construal (also see Yan 2022). Applying the diagnostics used in (25)/(26) to the baseline example (22), it is shown in (29) that an unambiguously eventive *say* leads to the high reading rather than the low one. This leads to the generalization that the high construal of adverbial *when* is licensed by eventive *say*.

- (29) I saw Puffy in Canary Wharf
- a. when she *enthusiastically/loudly said* that she would leave. (high ✓, low ✗)
 - b. when she *was saying* that she would leave. (high ✓, low ✗)

Interestingly, it is further demonstrated by (30) that the low construal of adverbial *when* is licensed by stative *say*. Imagine a context, in which Puffy is a famous TikTokker who always posts on her social platform to her fans when and where she will show up and shoot videos. Under such a context, the high construal in (30) is not as available as the low one.

- (30) I have never seen Puffy in Canary Wharf when her tweet *says* that she goes there. (high ✗, low ✓)

The inanimate subject *her tweet* indicates to some degree that (30) involves the stative use of *say*, i.e., the Source of Puffy's information (cf. (24b)). The stative status of *say* is further borne out by the fact that neither of the eventive diagnostics works out, as given in (31). Combined with what we have seen in (29), it is fair to conclude that the high construal of adverbial *when* is only allowed by eventive *say*, and its low construal is available only with stative *say*.

- (31) I have never seen Puffy in Canary Wharf
- a. *when her tweet *enthusiastically/loudly* says that she goes there.
 - b. *when her tweet *is saying* that she goes there.

The correlation between the interpretation of *when* and the structure of SAY verbs can now account for the disappearing low construal within the non-bridge context like (23) in a different fashion. Applying the differences between eventive and stative *say* as diagnostics to the non-bridge verb *exclaim* again, it is clear that *exclaim* is incompatible with the Source subject (32), and owns a series of eventive properties (33). These results suggest that *exclaim*, as a variety of SAY verbs, only has the eventive use, with the low construal of *when* unavailable.

- (32) *Puffy's tweet *exclaimed* she would leave at midnight. (*Source, eventive SAY)
- (33) a. Puffy *enthusiastically/loudly* *exclaimed* that she would leave.
 b. Puffy *was exclaiming* that she would leave.

In sum, I have shown that the ambiguous interpretations of adverbial *when* correlates with the different uses of SAY verbs: the high reading of *when* is available only with eventive SAY verbs, whereas its low reading is licensed by stative SAY verbs only. As will be seen soon, this correlation, together with the two different structures of *say* already discussed in Sect. 3.1, will inspire the core of my proposed analysis which does not appeal to any kind of operator movement.

3.3 Proposal: the event argument and its Agree relation with *when*

In what follows I flesh out what I call an Agree-based approach to Geis-ambiguities, which consists of two crucial assumptions. First, I postulate that a Davidsonian event argument (see Davidson 1967) is located in the specifier of the highest VP-shell of eventive SAY verbs (cf. Kratzer 1995, 1996, Stowell 1996, 2007). As illustrated in (27), VoiceP is the highest VP-shell within the entire structure of eventive *say*, whose specifier position is already taken up by the Agent argument. Here I simply assume that the VoiceP has multiple specifier positions, and that the Davidsonian event argument (represented by 'e') is placed in the outer Spec,VoiceP, as schematized in (34).

- (34) [_{VoiceP} e [_{VoiceP} Agent [_{Voice'} Voice [_{vP} v_{do} SayP]]]]

Recall that the structure of stative *say*, as illustrated in (28), does not involve the same VoiceP as its eventive counterpart does. The highest VP-shell in that configuration is just the vP headed by v_{be}, meaning that no Davidsonian event argument is assumed to exist there. In such a situation, two obvious possibilities arise: either stative *say* (as a representative of SAY verbs) does not introduce any ontological entities at all, or it just refers to a different one (see a similar line of thinking in Maienborn 2005:300). For the following discussion I adopt the latter option, but remain agnostic as to what exactly this different entity is, since my

proposed analysis will not be affected regardless of the identity of the entity, as will be clearer soon.⁷

My second assumption is that there is a syntactic dependency between adverbial *when* and the event argument, which involves constructing a predicate in the semantics (cf. Hall and Caponigro 2010). In other words, this syntactic dependency needs to be interpreted as predicate abstraction at LF. To implement such a syntax-semantics connection, I borrow the insights from Adger and Ramchand (2005)’s feature valuation system, which consists of a pair of syntactic features, namely, Λ and ID . The syntactic formative that bears the Λ -feature is interpreted as the predicate abstractor at the interface, and the ID -feature marks the variable position that is abstracted over. For any syntactic object that bears these two syntactic features, its correspondence with the LF interface allows it to be interpreted as predicate abstraction, as schematized in (35).

$$(35) \quad [\Lambda \dots \text{ID}] \rightarrow \lambda x \dots x$$

(Adger and Ramchand 2005:173, (38))

As to the ambiguous temporal adverbial clauses in question, I posit that adverbial *when*, as a base-generated complementizer, first bears the Λ -feature, which ensures that it can be interpreted as a predicate abstractor at the LF interface. Since *when* is also the element that receives ambiguous interpretations, it bears another unvalued ID -feature which must receive a value via Agree during the derivation. The matching valued ID -feature that the unvalued ID -feature on adverbial *when* will Agree with, I postulate, is borne only by the event argument introduced by the eventive *SAY* verbs. For the sake of completeness, I propose that *zeit* (meaning ‘time’ in German) is the value of the ID -feature on the event argument, lining up with Davidson (1967)’s idea that the Davidsonian event argument is a spatiotemporal entity in the world. The schema in (36) demonstrates how the Agree operation takes place in syntax first, and later allows the temporal adverbial clause involved to be interpreted as predicate abstraction at LF (‘*t*’ is used to represent the temporal variable conventionally).

$$(36) \quad \text{when}[\Lambda, \text{ID}:] \dots e[\text{ID}:\text{zeit}] \rightarrow \text{when}[\Lambda, \text{ID}:\text{zeit}] \dots e[\text{ID}:\text{zeit}] \rightarrow \lambda t \dots t$$

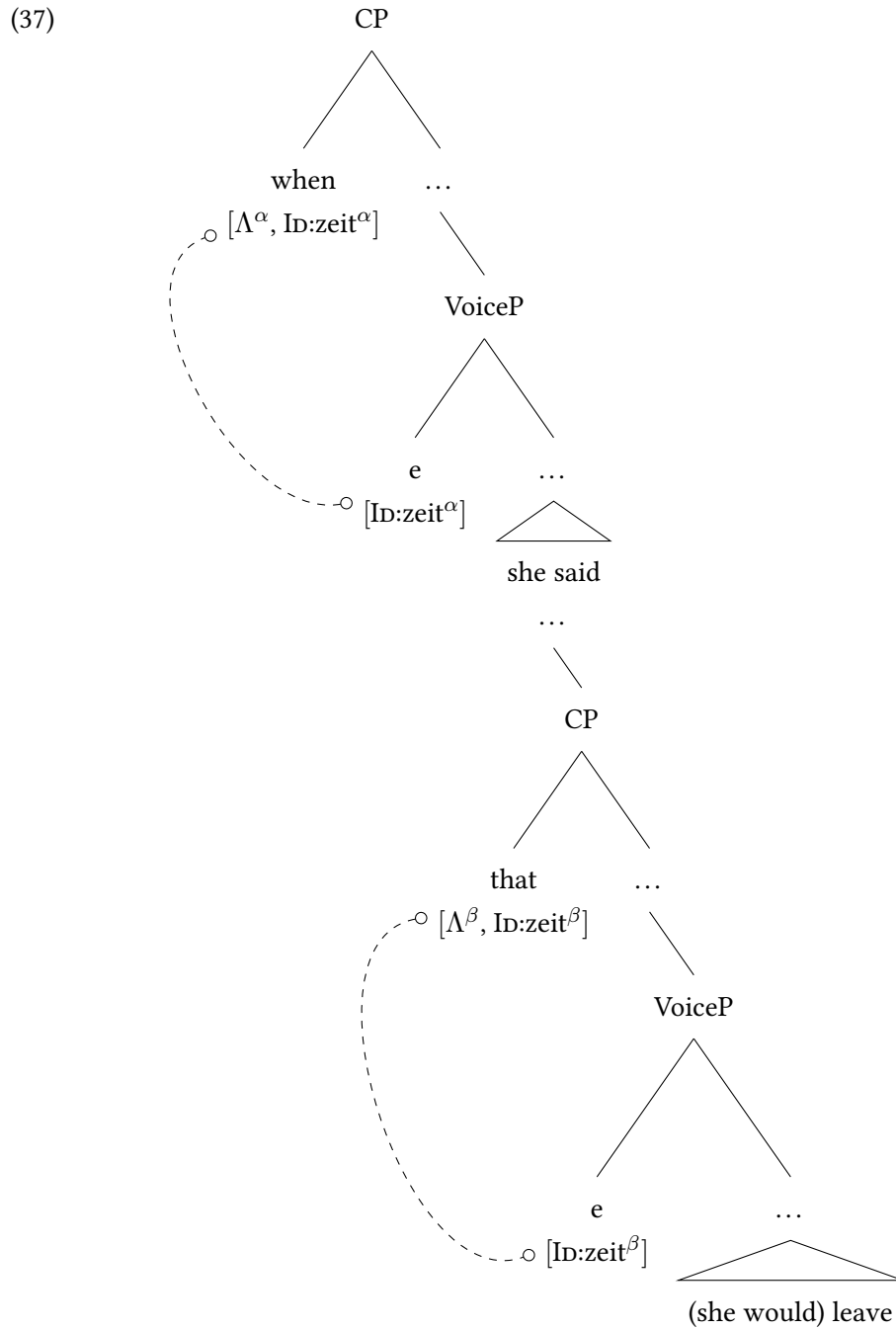
3.4 Implementation

With the preliminaries in mind, I now show two separate derivations behind Geis-ambiguities via the feature valuation system just outlined. Let me start off with the high construal of adverbial *when* that correlates with the eventive use of *say*, the derivation of which is illustrated in (37).

The first chunk that is built up during the derivation is the lower, embedded clause, which contains the embedded predicate *leave* and the intermediate complementizer *that*. Perceiving *leave* as an eventive predicate, I assume that an event argument is located in its Spec,VoiceP and bears the valued feature $[\text{ID}:\text{zeit}]$. With respect to the featural specification of *that*, I presume that it is not fundamentally different from that

⁷ A potential candidate for this entity introduced by stative *say* or *SAY* verbs is the Kimian state argument (see Kim 1969, 1976), as argued by Maienborn (2005). Unlike the Davidsonian event argument which is defined as a spatiotemporal entity, the Kimian state argument is an abstract object that denotes a property being instantiated at a particular time (Maienborn 2005:303, (47), Rothmayr 2009:7). I refer interested readers to Maienborn (2005) and Rothmayr (2009) for more detailed discussion of this state argument.

of adverbial *when*, given that both of them have the same function of introducing clauses. This means the intermediate complementizer bears a Λ -feature plus an unvalued Id-feature (i.e., $[\Lambda, \text{Id:}]$). The latter feature on the complementizer is required to be valued during the derivation, otherwise the derivation would crash. Thus in the midst of constructing the lower clause, the unvalued Id-feature on *that* undergoes Agree with the valued Id-feature on the event argument in its c-command domain and receives the zeit-value. For the sake of clarity, this Agree chain within the lower clause is indicated by the superscript β .



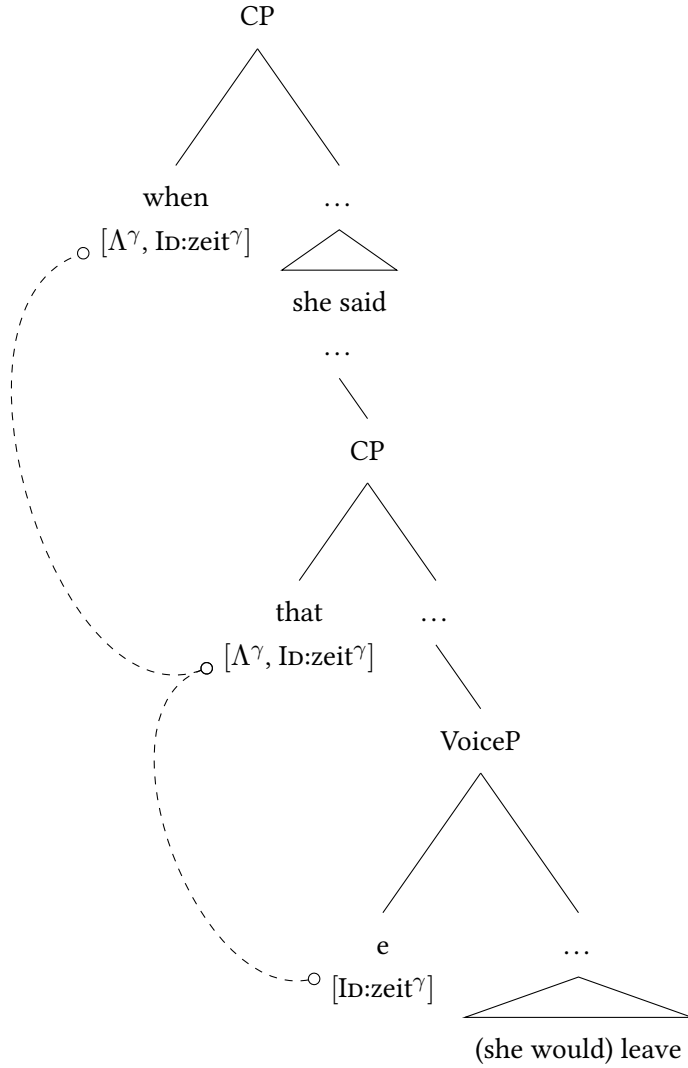
As the derivation proceeds, the event argument that stems from eventive *say* and adverbial *when* are sequentially involved in the construction of the higher clause. As with their respective counterparts in the lower clause, the event argument bears the valued ID-feature, and adverbial *when* is featurally specified as $[\Lambda, \text{ID:}]$. The same situation happens again in the higher clause: the unvalued ID-feature on adverbial *when* must undergo Agree with a syntactic formative that bears the matching ID-feature in its c-command domain to receive the value; otherwise, the derivation would crash. In this configuration, the closest goal for the unvalued ID-feature on *when* is the feature $[\text{ID:zeit}]$ on the event argument that is contained within the outmost VP-shell of eventive *say*. To be distinguishable from the Agree chain β that is already established in the lower clause, the current one is indicated by the superscript α . A simpler illustration of these two independent Agree chains is given as below.

(38) $\text{when}[\Lambda^\alpha, \text{ID:zeit}^\alpha] \dots e[\text{ID:zeit}^\alpha] \dots \text{that}[\Lambda^\beta, \text{ID:zeit}^\beta] \dots e[\text{ID:zeit}^\beta]$

It is clear from (38) that adverbial *when* only interacts with the event argument in the higher clause. The *wh*-complementizer has nothing to do with either the intermediate complementizer *that* or the embedded event argument. That the Agree chain α is established only within the higher clause, thus, provides a direct account for the correlation between the high construal of *when* and eventive *say*. Note that the configuration in (37)/(38) can also be applied to temporal adverbial clauses that contain other eventive SAY verbs like *exclaim* (e.g., 23), thereby explaining why the low construal of *when* is missing. For the sake of space, I will not illustrate the structure of *exclaim* here (as a reference, see the structure that involves the predicate *scream* in Major 2021:57, (98)).

Next, we can move on to the derivation of low construal of adverbial *when* in (39), which I haven't shown in Sect. 3.2 has correlation with the stative use of *say*. Identical to the derivation in (37), the lower, embedded clause in this scenario is still the first part of the entire temporal adverbial clause that is built up during the derivation. In the meantime, the intermediate complementizer *that* has to Agree with the event argument that is related to the eventive predicate *leave*, ensuring that the unvalued ID-feature on the former receives the zeit-value from the matching feature on the latter.

(39)



The crucial difference arises as the derivation moves forwards into the higher clause, where no event argument is assumed to exist. This is due to the reason that the predicate in the higher clause is stative *say*, which differs from its eventive counterpart in lacking both the Davidsonian event argument and the maximal projection of Voice. The consequence is that the *Id*-feature on adverbial *when* would not be valued locally, and the *wh*-complementizer has to probe further into its c-command domain to look for the matching feature. At this point of derivation, the *Id*-feature on the intermediate complementizer *that* has already been valued, which renders it the closet goal within the c-command domain of adverbial *when*. Thus, to avoid the potential derivational crash, the unvalued *Id*-feature on adverbial *when* must undergo Agree with the already-valued *Id*-feature on the intermediate complementizer *that*. This derivation leads to the successive-cyclic Agree chain indicated by the superscript γ , which is simply illustrated in (40).

(40) $\text{when}[\Lambda^\gamma, \text{Id:zeit}^\gamma] \dots \text{that}[\Lambda^\gamma, \text{Id:zeit}^\gamma] \dots e[\text{Id:zeit}^\gamma]$

Comparing with (38) where two Agree chains α and β are independent from each other, there is only

one single Agree chain γ in (40) that connects adverbial *when* with both the intermediate complementizer and the embedded event argument. This results in the long-distance interpretation of the topmost *wh*-element (i.e., the low construal of *when*), which is realised not by successive-cyclic operator movement (e.g., (2)), but by successive-cyclic Agree chain instead.

4 Conclusion

In this paper, I have argued against the traditional operator movement approach to Geis-ambiguities, and instead proposed an alternative, Agree-based approach to the same ambiguous effects. My proposal has captured two crucial empirical facts that have escaped attention in the previous literature. One is that adverbial *when* is syntactically distinct from canonical *wh*-operators, which I have argued is actually a *wh*-complementizer base-generated in the C head. The other is that the ambiguous interpretations of *when* are actually reflections of different uses of local verbs (e.g., SAY verbs) in temporal adverbial clauses. The key prediction from my proposed analysis here is that, long-distance dependency is generally predicted to be blocked except for certain contexts where local verbs are able to offer some ‘bridges’. By contrast, for those analyses based on operator movement, long-distance dependency is always predicted to be possible in contexts where no canonical islands show up. If not, then the dependency must be blocked by some additional principle other than islands. It is so far unknown to me to what extent my non-operator movement approach can account for other cases of Geis-ambiguities that do not involve *wh*-elements. I will have to leave this question for future research.

References

- Adger, David, and Gillian Ramchand. 2005. Merge and move: *Wh*-dependencies revisited. *Linguistic Inquiry* 36:161–193.
- Bresnan, Joan, and Jane Grimshaw. 1978. The syntax of free relatives in English. *Linguistic Inquiry* 9:331–391.
- Chierchia, Gennaro. 1993. Questions with quantifiers. *Natural Language Semantics* 1:181–234.
- Citko, Barbara. 2002. (Anti)reconstruction effects in free relatives: A new argument against the comp account. *Linguistic Inquiry* 33:507–511.
- Davidson, Donald. 1967. The logical form of action sentences. In *The logic of decision and action*, ed. by Nicholas Rescher, 81–95. Pittsburgh: Pittsburgh University Press.
- Erteschik-Shir, Nomi. 1973. On the nature of island constraints. Doctoral dissertation, MIT.
- Folli, Raffaella, and Heidi Harley. 2005. Flavors of *v*: Consuming results in Italian and English. In *Aspectual inquiries*, ed. by Paula Kempchinsky and Roumyana Slabakova, 95–120. Dordrecht: Springer.
- Geis, Michael. 1970. Adverbial subordinate clauses in English. Doctoral dissertation, MIT.

- Grimshaw, Jane. 2015. The light verbs *say* and *SAY*. In *Structures in the mind: Essays on language, music, and cognition in honor of Ray Jackendoff*, ed. by Ida Toivonen, Piroska Csúri, and Emile van der Zee, 79–99. Cambridge, MA: MIT Press.
- Haegeman, Liliane. 2007. Operator movement and topicalisation in adverbial clauses. *Folia Linguistica* 41:279–325.
- Haegeman, Liliane. 2009. The movement analysis of temporal adverbial clauses. *English Language and Linguistics* 13:385–408.
- Haegeman, Liliane. 2010. The internal syntax of adverbial clauses. *Lingua* 120:628–648.
- Haegeman, Liliane. 2012. *Adverbial clauses, main clause phenomena, and the composition of the left periphery*. The Cartography of Syntactic Structures, Volume 8. Oxford: Oxford University Press.
- Hall, David P., and Ivano Caponigro. 2010. On the semantics of temporal *when*-clauses. In *Proceedings of Semantics and Linguistic Theory 20*, ed. by Nan Li and David Lutz, 544–563. Ithaca, NY: CLC Publications.
- Kim, Jaegwon. 1969. Events and their descriptions: Some considerations. In *Essays in the Honor of Carl G. Hempel*, ed. by Nicholas Rescher, 198–215. Dordrecht: Reidel.
- Kim, Jaegwon. 1976. Events as property exemplifications. In *Action theory*, ed. by Myles Brand and Douglas Walton, 159–177. Dordrecht: Reidel.
- Kratzer, Angelika. 1995. Stage-level and individual-level predicates. In *The generic book*, ed. by Gregory N. Carlson and Francis Jeffry Pelletier, 125–175. Chicago: University of Chicago Press.
- Kratzer, Angelika. 1996. Severing the external argument from its verb. In *Phrase structure and the lexicon*, ed. by Johan Rooryck and Laurie Zaring, 109–137. Dordrecht: Kluwer.
- Larson, Richard K. 1987. “Missing prepositions” and the analysis of English free relative clauses. *Linguistic Inquiry* 18:239–266.
- Larson, Richard K. 1990. Extraction and multiple selection in PP. *The Linguistic Review* 7:169–182.
- Maienborn, Claudia. 2005. On the limits of the Davidsonian approach: The case of copula sentences. *Theoretical Linguistics* 31:275–316.
- Major, Travis. 2021. On the nature of “say” complementation. Doctoral dissertation, UCLA.
- May, Robert. 1985. *Logical form: Its structure and derivation*. Cambridge, MA: MIT Press.
- May, Robert. 1988. Ambiguities of quantification and *wh*: A reply to Williams. *Linguistic Inquiry* 19:118–135.
- McCloskey, James. 2000. Quantifier float and *wh*-movement in an Irish English. *Linguistic Inquiry* 31:57–84.

- Pesetsky, David. 1987. *Wh-in-situ: Movement and unselective binding*. In *The representation of (in)definiteness*, ed. by Eric J. Reuland and Alice G. B. ter Meulen, 98–129. Cambridge, MA: MIT Press.
- Pylkkänen, Liina. 2008. *Introducing arguments*. Cambridge, MA: MIT Press.
- Rothmayr, Antonia. 2009. *The structure of stative verbs*. Amsterdam: John Benjamins.
- Stowell, Tim. 1996. The phrase structure of tense. In *Phrase structure and the lexicon*, ed. by Johan Rooryck and Laurie Zaring, 277–291. Dordrecht: Kluwer.
- Stowell, Tim. 2007. The syntactic expression of tense. *Lingua* 117:437–463.
- Williams, Edwin. 1988. Is LF distinct from S-Structure? A reply to May. *Linguistic Inquiry* 19:135–146.
- Wu, Danfeng. 2022. Why *if or not but ✓whether or not. *Linguistic Inquiry* 53:412–425.
- Xiang, Yimei. 2023. Quantifying into *wh*-dependencies: Multiple-*wh* questions and questions with a quantifier. *Linguistics and Philosophy* 46:429–482.
- Yan, Qiu hao Charles. 2022. The structure of SAY verbs and temporal modification. *Snippets* 43:7–9.